The University of the South Pacific

Serving the Cook Islands, Fiji, Kiribati, Marshall Islands, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, and Vanuatu.

School of Computing, Information and Mathematical Sciences

CS211: Computer Organization

Test 1 – Semester II, 2014

Time Allowed: 1 hour

Total Marks: 40 (15% of final grade)

Solution

Name: _____ **Student ID:** _____

INSTRUCTIONS:

- There are 8 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

a) Convert the binary number 11000011101010 into octal and hexadecimal? (2 marks)

1100001110101 0

$30EA_H$

(1)

11000011101010

30352_8

(1)

b) Convert the decimal number 600 to binary? (2 marks)

$\frac{600}{2}$		$\frac{75}{2}$		$\frac{9}{2}$		$\frac{1}{2}$	
$\frac{300}{2}$	0	$\frac{37}{2}$	1	$\frac{4}{2}$	1	0	1
$\frac{150}{2}$	0	$\frac{18}{2}$	1	$\frac{2}{2}$	0	∴	~~1000011000~~
75	0	9	0	1	0		1001011000

c) Express the decimal value -23 in 8 bit binary using two's complement? (2 marks)

$-23 \Rightarrow |-23| \Rightarrow 00010111$

$\Rightarrow 11101000$ (flip)

$+ \quad 1$ (Add 1)

11101001

d) Express the binary number 000000101 into decimal using Excess method? (2 marks)

$(000000101)_2 = x + 2^{9-1}$ → ~~Should have been 8 bits~~

$5 = x + 256$

$x = -251$

Question 2

Convert -15.3 to IEEE single-precision format. Represent the final answer in hexadecimal. (4 marks)

15 ⇒ 1111

0.3 × 2

0 | .6 × 2

1 | .2 × 2

0 | .4 × 2

0 | .8 × 2

1 | .6 × 2

1 | .2 × 2

0 | .4

15.3 = 1111.0100110011....

Normalise ⇒ $1.nnn \times 2^{exp}$

⇒ $1.1110100 1 \cdots \times 2^{3}$

Stored exp = actual + excess

= 3 + 127

= 130

⇒ 10000010

1	10000010	1110100110011....
sign	~~frac~~ exp	frac

C174CCCD

Question 3

Differentiate between instruction level and processor level parallelism? Give an example each. (3 marks)

Instruction level - parallelism is exploited within individual instructions to get more instructions per second out of the machine.
e.g pipelining

Processor-level - multiple CPUs work together to achieve parallelism
e.g multiprocessors, multi computer.

Question 4

You received an encoded Hamming code word 111011100001. You want to determine what the original data word sent was. Assume the received code word was encoded for one-bit error detection and correction using Hamming code with even parity. What was the original data word sent? Express your answer as a hexadecimal value. (3 marks)

111011100001

1 1 1 0 1 1 1 0 0 0 0 1

1 2 3 4 5 6 7 8 9 10 11 12

Bit 1: 1, 3, 5, 7, 9, 11 — 4 ✓

Bit 2: 2, 3, 6, 7, 10, 11 — 4 ✓

Bit 4: 4, 5, 6, 7, 12 — 4 ✓

Bit 8: 8, 9, 10, 11, 12. — 1 ✗

∴ error at Bit 8

Data word ⇒ 11110001

F1

Question 5

List the levels in the six level computer and show how each level is supported by the level below it? (5 marks)

Problem Oriented level

Translation (compiler)

Assembly level

Translation (assembler)

OS level

partial interpretation

ISA level

Interpretation or direct execution

microarchitecture level

Hardware

Digital logic

Question 6

A disk drive has 1000 cylinders, 8 tracks per cylinder, 32 sectors per track, and 512 bytes per sector. The rotation time is 4ms. The read/write head takes 8ms to move between adjacent cylinders and the maximum seek time is 300ms. A data file contains 300KB of data stored consecutively. What is the average time needed to read the data? (3 marks)

$$\#\text{ of tracks} = 300\,KB \times \frac{1024\,B}{1\,KB} \times \frac{1\,sect}{512\,B} \times \frac{1\,track}{32\,sect}$$

$$= \underline{\underline{18.75 \text{ tracks}}}$$

$$\text{Transfer time} = \#\text{ of tracks} \times RT$$

$$= 18.75 \times 4$$

$$= 75$$

$$\text{Avg time} = \text{avg seek} + \text{avg lat} + \text{avg transfer} + \text{avg s/w}$$

$$= \frac{300}{2} + \frac{4}{2} + 75 + \underline{\underline{18}}$$

$$= \underline{\underline{195 \text{ ms}}}$$

tracks	Probability	# of s	s/w time
1	1/8	2	16
2	1/8	2	16
3	⋮	2	16
4	⋮	2	16
5	⋮	2	16
6	⋮	2	16
7	⋮	3	24
8	1/8	3	24

Average s/w

$$\left(\frac{6}{8} \times 16\right) + \left(\frac{2}{8} \times 24\right)$$

$$= 12 + 6$$

$$= 18.$$

Question 7

Write the output for the following machine language program if the inputs are 5, 10, 15, 20 and -1. The inputs are in sequential order. Show the accumulator value for each input and any changes to it done by the program. (4 marks)

```
0011000000000000   3  0
1011000000001010   11 10
1000000000001110   8  14
0010000000001110   2  14
1001000000000000   9  0
0100000000000000   4  0
0000000000000000   0  0
0000000000000001   0  1
```

ACC = 5, 5/1, 10, 10/5, 15, 15/2, 20, 20/4, -1

Output = -1

Question 8 (10 marks)

A function is defined as follows:

$$f(n) = \begin{cases} n^3 & \text{for } n < 0 \\ 10 & \text{for } n = 0 \\ n^3 - 2n + 6 & \text{for } n > 0 \end{cases}$$

Write Wombat1 machine language program that reads an input integer value n, and computes and displays the output of the above function f(n). Write your program in the decimal version beginning with the memory/line address for each line.

```
0:  3  0  ; read n
2:  11 24 ; if n<0 jump to loc 24
4:  10 30 ; if n=0 jump to loc 30
6:  2  38 ; store n
8:  7  42 ; 2n
10: 5  44 ; 2n+6
12: 2  36 ; store f(n)
14: 1  38 ; load n
16: 7  38 ; nxn
18: 7  38 ; nxnxn
20: 6  36 ; n^3 - 2n+6
22: 9  34 ; jump to write
24: 7  38 ; nxn
26: 7  38 ; nxnxn
28: 9  32 ; jmp to write
30: 1  40 ; load 10
32: 4  0  ; write
34: 0  0  ; stop
36: 0  0  ; storage for f(n)
38: 0  0  ; storage for n
40: 0  10 ; storage for const 10
42: 0  2  ; "  "  const 2
44: 0  6  ; "  "  "  6
```

[EXTRA PAGE]